EDUCATIONAL MACHINE LEARNING PROJECT | P21

Tumour Tagger

Breast Cancer Classification using Logistic Regression

Supervised learning | EDA | Feature selection | Model evaluation

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PROGRAMME **BCA, LJ University**	DATASET **ml_data_p21.csv**

Educational project - not a medical diagnostic tool.

1 Problem Statement

This project uses supervised machine learning to classify breast tumour records as **benign (B)** or **malignant (M)** from cell-nucleus measurements. The goal is to predict the diagnosis label and evaluate the classifier, with special attention to **malignant recall**: a false negative means a malignant-labelled record predicted as benign. This is an educational project, not a medical diagnostic tool.

Objectives: inspect and clean the CSV; explore class balance, distributions and correlations; split data before fitting preprocessing to reduce leakage; compare Logistic Regression settings; evaluate with multiple metrics.

Setup: data from **ml_data_p21.csv**; stratified **80/20** train-test split, **random_state=42**.

2 Dataset Description and Data Dictionary

The CSV has **569 rows and 33 columns** before cleaning: **357 benign (62.7%)** and **212 malignant (37.3%)** records, with 0 duplicate rows. The ID column and the fully empty *Unnamed: 32* column were removed, leaving 569 rows and 32 columns including the engineered feature.

Field / group	Meaning
diagnosis	Target label: B = benign; M = malignant.
radius_mean, texture_mean, perimeter_mean, area_mean	Mean size and texture measurements.
smoothness_mean, compactness_mean, concavity_mean, concave points_mean, symmetry_mean, fractal_dimension_mean	Mean shape, boundary and texture-related measurements.
radius_se, texture_se, perimeter_se, area_se	Standard-error values for corresponding measurements.
smoothness_se, compactness_se, concavity_se, concave points_se, symmetry_se, fractal_dimension_se	Standard errors for shape and texture-related measurements.
radius_worst, texture_worst, perimeter_worst, area_worst	Worst/largest values for corresponding measurements.
smoothness_worst, compactness_worst, concavity_worst, concave points_worst, symmetry_worst, fractal_dimension_worst	Worst-value shape and texture-related measurements.
id (removed)	Record identifier; excluded from modelling.
Unnamed: 32 (removed)	Empty extra column; excluded from modelling.
area_perimeter_ratio_mean	Engineered feature: area_mean divided by perimeter_mean.

Feature names follow the supplied dataset; descriptions are based on those names.

3 Cleaning and Exploratory Data Analysis (EDA)

Cleaning: removed the identifier and empty column; excluded rows without a target; filled any remaining missing inputs with training-set medians; added *area_perimeter_ratio_mean* = area_mean / perimeter_mean.

EDA was used to inspect class balance, compare a representative feature across classes, and assess relationships between size-related measurements.

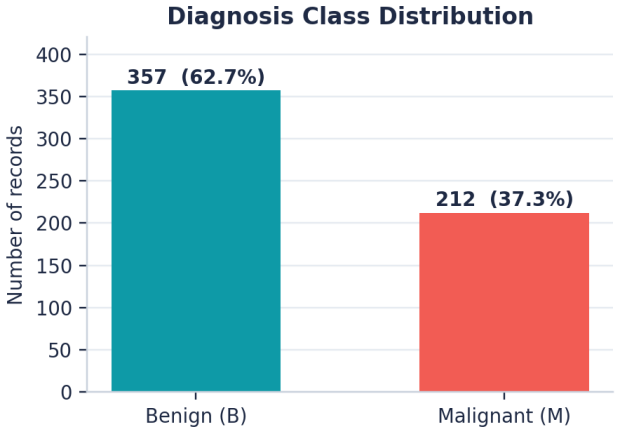


Figure 1. Diagnosis class counts.

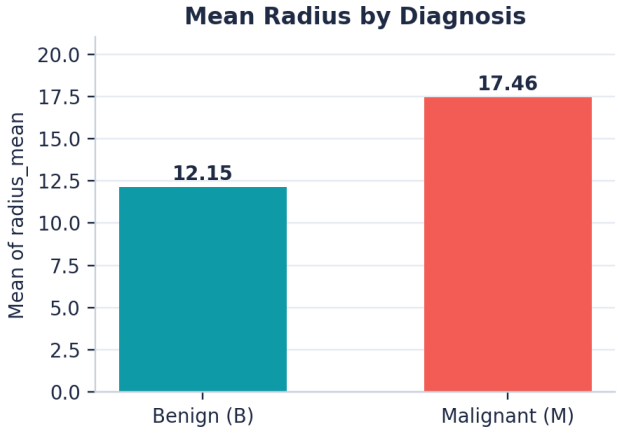


Figure 2. Mean of radius_mean by diagnosis.

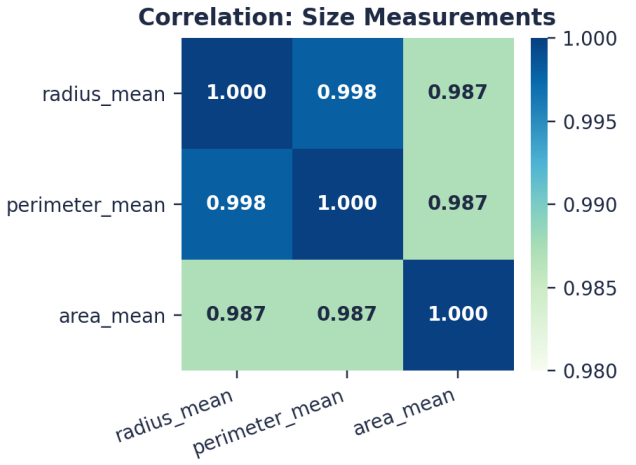


Figure 3. Correlation among selected size-related measurements. Correlation shows association, not causation.

Five EDA insights

1	Benign records outnumber malignant records (357 vs 212).
2	Numeric features have different ranges, motivating scaling.
3	Mean radius differs by class: benign mean **12.15**; malignant mean **17.46**.
4	Mean radius and mean perimeter are highly correlated (r = **0.998**).
5	Mean area is also strongly correlated with mean radius (r = **0.987**), suggesting overlapping information.

4 Model, Settings Tried and Comparison

CLEAN

Removed the identifier and empty column; excluded rows without a target; filled remaining missing inputs using training-set medians. Added *area_perimeter_ratio_mean*.

SPLIT

Stratified 80/20 split: **455** training and **114** testing records (random_state=42). Imputation, selection and scaling fitted on training data only.

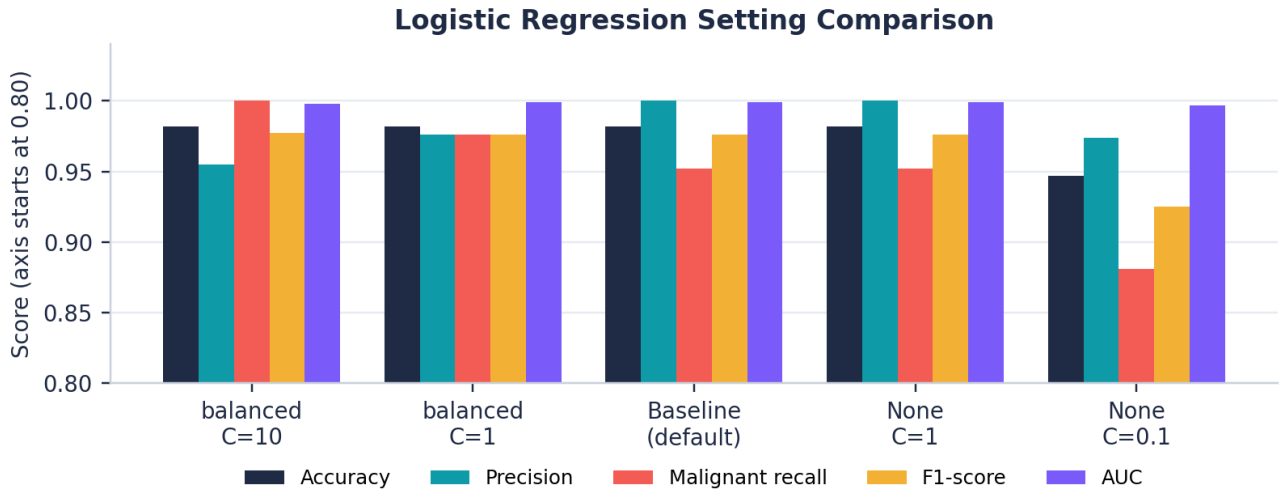
MODEL

SelectKBest (ANOVA F-test) kept **15 features**; StandardScaler applied before Logistic Regression. Five settings compared by regularisation (C) and class weighting.

Held-out test-set comparison

Setting	Accuracy	Precision	M recall	F1	Specificity	AUC
class_weight=balanced, C=10.0	**0.982**	**0.955**	**1.000**	**0.977**	**0.972**	**0.998**
class_weight=balanced, C=1.0	0.982	0.976	0.976	0.976	0.986	0.999
Baseline (default class weight)	0.982	1.000	0.952	0.976	1.000	0.999
class_weight=None, C=1.0	0.982	1.000	0.952	0.976	1.000	0.999
class_weight=None, C=0.1	0.947	0.974	0.881	0.925	0.986	0.997

Highlighted row = selected model.



Selection rule: highest malignant recall, with AUC as tie-breaker. **Selected setting: class_weight=balanced, C=10.0.** This prioritises fewer missed malignant-labelled test records, but does not establish clinical usefulness.

5 Results: Final Model Evaluation

The selected model (**class_weight=balanced, C=10.0**) was evaluated on 114 held-out test records. It classified **112 records correctly** and made **2 errors**.

98.25%	100%	95.45%	0.9977
ACCURACY	MALIGNANT RECALL	PRECISION	ROC-AUC

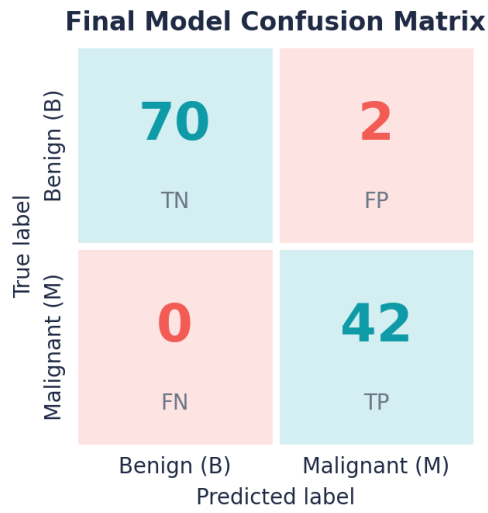


Figure 4. Confusion matrix.

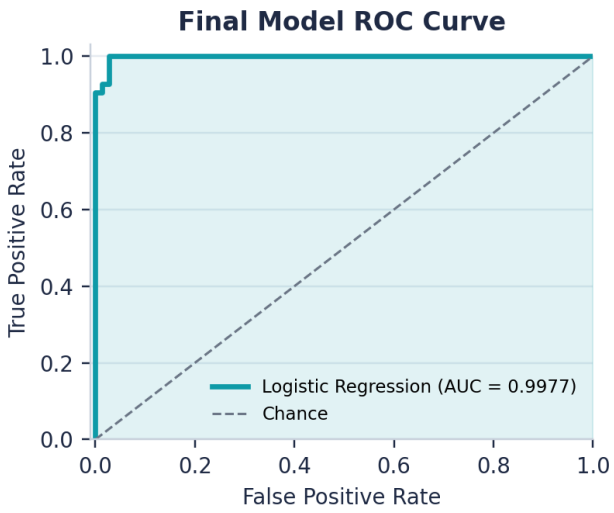


Figure 5. ROC curve (stepped curve drawn from the confusion-matrix counts).

Metric	Result	Meaning
Accuracy	**98.25%**	Share of all test records classified correctly.
Precision (malignant)	**95.45%**	Share of predicted malignant records that were actually malignant.
Malignant recall	**100.00%**	Share of actual malignant records identified.
F1-score	**0.977**	Harmonic mean of precision and recall.
Specificity	**97.22%**	Share of actual benign records correctly identified.
ROC-AUC	**0.9977**	Ranking separation across probability thresholds.

Confusion matrix counts: TN = 70, FP = 2, FN = 0, TP = 42. The model missed 0 malignant-labelled records and incorrectly labelled 2 benign records as malignant in this test split.

These results come from one fixed split and may vary with different data splits or modelling choices.

6 Conclusion and What I Learned

The project demonstrates a supervised ML workflow from cleaning and EDA through feature selection and Logistic Regression evaluation. The selected setting (**class_weight=balanced, C=10.0**) achieved **98.25% accuracy**, **100.00% malignant recall** and **0.9977 ROC-AUC** on the held-out test set. These results apply to this split only and are not evidence of a validated clinical model.

What I learned

- **Avoid data leakage:** split first, then fit imputation, feature selection and scaling on training data only.
- **Accuracy is not enough:** when missing a malignant case is costly, recall matters more; accuracy was 0.982 for four settings, but recall differed (1.000 vs 0.952).
- **Settings change the trade-off:** class_weight=balanced raised malignant recall, while a strong penalty (C=0.1) lowered it to 0.881.
- **Scaling matters:** features had very different ranges, so StandardScaler was needed before Logistic Regression.
- **Correlated features overlap:** radius, perimeter and area (r up to 0.998) carry similar information, so feature selection helps.

Illustrative predictions (synthetic)

Created by slightly perturbing selected held-out records, following the notebook's procedure. They are teaching examples, not patient-specific predictions.

Example	Predicted label	Estimated malignant probability
Illustrative input 1	B (benign)	**0.015**
Illustrative input 2	M (malignant)	**1.000**
Illustrative input 3	B (benign)	**0.026**

Limitations

- Educational project, not a medical diagnostic system.
- One train-test split only; no cross-validation or external validation (next step: cross-validation).
- Feature selection and threshold choices affect the balance of false positives and false negatives.
- Any real clinical use would need expert review and extensive independent validation.